

YUVAINTERN & NSDC SKILLING PROGRAM

CRYPTOGRAPHY & DATA PROTECTION

Comprehensive Internship Final Project Presentation

Candidate Name: Ansh Patel | **Role:** Cryptography & Data Protection Specialist
Organization: YuvaIntern (in partnership with NSDC & Henry Harvin Education)
Certificate Number: YI/2026/208210/451475 | **Date of Issue:** November 05, 2026

Core Project Objectives & Focus Areas

1. Cryptographic Fundamentals

Understand AES, RSA, ECC, and SHA-256 mathematical structures and application trade-offs.

2. Symmetric vs Asymmetric

Analyze performance, key distribution, and authentication mechanics across encryption models.

3. Practical Lab Implementations

Build hands-on cryptographic routines using Python, PyCryptodome, and secure password hashing (Argon2id).

4. KMS & Key Management

Evaluate enterprise key lifecycle, Hardware Security Modules (HSM), key rotation, and secret storage.

5. Codebase Audit & Hardening

Review legacy applications, identify weak ciphers (ECB, MD5), and remediate security vulnerabilities.

6. Capstone Architecture

Design and document an end-to-end multi-tier data protection vault for enterprise data.

6-Week Program Structure & Progress Roadmap

Week 1

Concepts

Submission

#129093

AES, RSA, ECC &
Hashing
fundamentals.

Week 2

Hands-On Lab

Submission

#129095

Python,
PyCryptodome,
AES-GCM,
Argon2id.

Week 3

Case Study

Submission

#129097

Data
classification, TLS
1.3 & KMS
lifecycle.

Week 4

Tool Mastery

Submission

#129098

Code audit, ECB
weakness, secret
exposure.

Week 5

Architecture

Submission

#129099

Envelope
encryption design
& threat models.

Week 6

Capstone

Submission

Completed

Full-scale secure
data protection
vault.

Cryptographic Tech Stack & Library Specifications

Python 3.11+ & PyCryptodome — Core Cryptographic Engine

Execution of AES-256-GCM, RSA-4096, and ECC secp256r1 algorithm implementations.

Argon2id & bcrypt — Password Hashing Standards

Memory-hardened key derivation functions resistant to GPU and ASIC acceleration attacks.

OpenSSL & X.509 — PKI & Certificate Tools

Digital certificate generation, Certificate Authority signing, and TLS channel validation.

HashiCorp Vault / AWS KMS — Key Management Systems

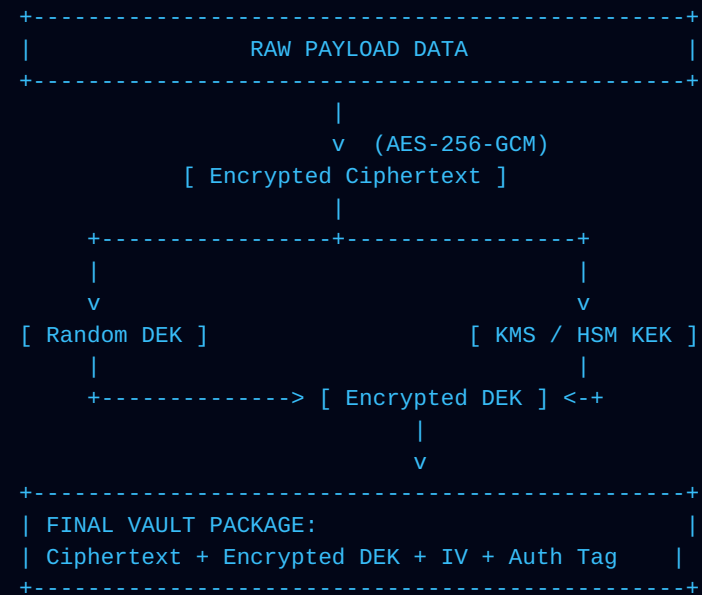
Envelope encryption, HSM storage, automated key rotation, and access control policies.

System Architecture: Envelope Encryption & KMS Workflow

Envelope Encryption Mechanism

- **Data Encryption Key (DEK):** Generated dynamically per payload using CSPRNG for high-speed local symmetric encryption (AES-256-GCM).
- **Key Encryption Key (KEK):** Stored securely inside a Hardware Security Module (HSM) or HashiCorp Vault.
- **Key Wrapping:** The KMS encrypts the local DEK using the KEK before storing it alongside the ciphertext package.
- **Decryption Flow:** Application sends encrypted DEK to KMS -> KMS decrypts DEK -> Local DEK decrypts payload.

Architectural Diagram Representation



Cryptographic Code Audit & Vulnerability Remediation

[CRITICAL] AES Block Cipher Mode

Weakness: AES-ECB mode was used, leaking visual structural patterns in encrypted data.

Remediation: Migrated to AES-256-GCM with unique 96-bit IVs and 128-bit authentication tags.

[HIGH] Password Storage

Weakness: Legacy systems stored passwords using insecure MD5 / plain SHA-256 digests.

Remediation: Implemented Argon2id key derivation with per-user salt and memory hardening.

[CRITICAL] Secret Key Exposure

Weakness: RSA private keys and API credentials were hardcoded directly in code.

Remediation: Externalized key storage into HashiCorp Vault with dynamic environment loading.

[MEDIUM] Initialization Vector

Weakness: Ciphers utilized static or predictable IV counters during packet encryption.

Remediation: Enforced cryptographically secure random IV generation (CSPRNG) per message.

Program Completion & Official Internship Certification

OFFICIAL CERTIFICATE OF EXPERIENCE

This is to certify that **Ansh Patel** successfully completed an internship with **YuvaIntern** in the role of **Cryptography & Data Protection Specialist**.

Key Internship Verification Records:

- **Certificate Number:** YI/2026/208210/451475
- **Date of Issue:** November 05, 2026
- **Partner Organization:** NSDC (National Skill Development Corporation)
- **Executive Signatory:** Kounal Gupta (Founder, YuvaIntern.com)
- **Affiliated Institution:** Henry Harvin Education